

MAVMRF Solution Brief

Solicitation Alignment

****Program:**** DIU Robotic Exclusion & Engagement Framework (REEF)

****Solution Component:**** Component 1 - Detect, Track, and Classify

Executive Summary

Maritime Autonomous Vehicle Monitoring and Response Framework (MAVMRF) is a modular, COTS-oriented software framework for maritime monitoring and operator decision support. The system delivers an end-to-end detect-track-classify pipeline using multi-sensor inputs (sonar, acoustic, optical, magnetic), AI-assisted classification, sensor fusion, and persistent tracking to produce actionable operator outputs in real time.

MAVMRF is structured for rapid transition from prototype evaluation to platform integration through a clean subsystem architecture and an open adapter interface for external sensor feeds.

Operational Problem Addressed

Ports, harbors, and critical waterways require reliable, scalable methods to detect and evaluate subsea contacts with sufficient decision time for human operators. Current approaches are often fragmented by sensor type or deployment model.

MAVMRF addresses this gap by unifying sensing, fusion, tracking, and classification outputs into a single operational pipeline with machine-readable reporting and visual evidence generation.

Technical Approach

MAVMRF executes the following workflow:

1. Multi-sensor ingestion (simulated live stream by default; file input optional)
2. Sensor preprocessing and normalization
3. Clutter/false-alarm filtering with operator-selectable sensitivity
4. AI-assisted object detection and classification
5. Weighted multi-sensor fusion into unified object state estimates
6. SORT-style multi-object tracking with persistent IDs
7. Rule-based advisory/report generation and visualization export

Component 1 Capability Mapping

- ****Detection:**** Provides per-contact detection information suitable for rapid operator interpretation.
- ****Track Estimation:**** Maintains confident, continuous tracks with trajectory history and estimated movement over time.
- ****Classification/Discrimination:**** Supports eight marine classes and report-level differentiation via ``contact_type`` (``biological_like``, ``non_biological_like``, ``unknown``).
- ****Clutter/False Alarm Reduction:**** Implements configurable filter sensitivity (``--filter-sensitivity``)

low|medium|high`).

- ****Multi-Sensor Coverage:**** Integrates sonar, acoustic, optical, and magnetic modalities in one pipeline.
- ****Open Architecture:**** Includes adapter contract for rapid integration of fixed and mobile sensor feeds.

Operator-Facing Outputs

Each report frame includes key decision-support fields:

- `bearing`
- `estimated\_range`
- `bearing\_rate`
- classification label and confidence
- track ID and track history
- contact differentiation (`contact\_type`)

Outputs are generated as structured JSON artifacts and time-indexed visualizations to support both machine processing and operator review.

Evidence Package

Representative early/mid/late artifacts are included in the submission package:

- `evidence/visualization\_frame\_00005.png`
- `evidence/visualization\_frame\_00020.png`
- `evidence/visualization\_frame\_00040.png`
- `evidence/report\_frame\_00005.json`
- `evidence/report\_frame\_00020.json`
- `evidence/report\_frame\_00040.json`

Technical Maturity and Transition Path

MAVMRF is submission-ready as a working Component 1 prototype and can be advanced rapidly through:

1. Live sensor integration through adapter implementations.
2. Parameter calibration against representative harbor/coastal datasets.
3. Edge hardware deployment and sustained runtime benchmarking.
4. Expanded evaluation in varied environmental and traffic conditions.

Summary

MAVMRF provides a credible, modular, and integration-ready Component 1 solution with demonstrated detect-track-classify functionality, operator-relevant metrics, clutter filtering control, biological/non-biological differentiation support, and open-architecture extensibility for fixed and mobile deployments.